

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir, get_model_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)

# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```
# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
=====
CONFIGURACIÓN INICIAL
=====
Configuración Valor
      Device  cuda
      Seed    42
Output Width  100
=====
```

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'p-annealing',        # P-annealing (Lp^p con p decreciente)
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 12500,                 # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', # Path base para cl
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments
    'extras': {
        'expansion': 16,
        'sparsity_coef': 3e-1,
        'epochs': 200,
        'early_stopping': False,
        'learning_rate': 1e-3,
        'p_start': 1.0,
```

```

        'p_end': 0.2,
        'n_sparsity_updates': 10,
        'anneal_start': 5000,
        'batch_size': 2048,
        'warmup_steps': 1000,
        'sparsity_warmup_steps': 2000,
        'estimated_epochs': 200
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')

```

```

Metadata de naming:
variante: standard
tecnica: p-annealing
capa: 6
juegos: 12500
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 16, 'sparsity_coef': 0.3, 'epochs': 200, 'early_stopping': False, 'l

```

```

extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

print(f'extras_id: {extras_id}')

```

```

Nuevo extras registrado: ex3 → {'expansion': 16, 'sparsity_coef': 0.3, 'epochs': 200, 'earl
extras_id: ex3

```

3. Configuración del SAE

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```
# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 8192,
    'tied_weights': False,

    # Entrenamiento
    'batch_size': 2048,
    'num_epochs': 200,
    'learning_rate': 1e-3,
    'warmup_steps': 1000,          # steps de LR warmup lineal
    'sparsity_warmup_steps': 2000, # steps de rampa de sparsity (0→1)
    'sparsity_coef': 3e-1,        # alpha inicial (loss sum(dim=-1).mean())

    # P-annealing
    'p_start': 1.0,               # p inicial (equivalente a L1)
    'p_end': 0.2,                 # p final
    'n_sparsity_updates': 10,     # número de saltos discretos
    'anneal_start': 5000,         # step en que empieza el annealing (después del warmup)
    'estimated_epochs': 200,      # épocas estimadas para condensar el schedule

    # Early Stopping
    'early_stopping': False,
    'patience': 50,              # (inactivo: early_stopping=False)

    # Datos
    'activations_file': Path(f"../../activations/data/layer{NAMING_META['capa']}_{NAMING_META['juegos']}.npy"),
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}
```

```

# Crear directorio local del modelo
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}

for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['batch_size', 'num_epochs', 'learning_rate', 'warmup_steps', 'sparsity_warmup_steps']:
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['p_start', 'p_end', 'n_sparsity_updates', 'anneal_start', 'estimated_epochs']:
    config_data['Categoría'].append('P-Annealing')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)

```

===== CONFIGURACIÓN DEL SAE =====

Categoría	Parámetro	Valor
Arquitectura	input_dim	512
Arquitectura	hidden_dim	8192
Arquitectura	tied_weights	False
Entrenamiento	batch_size	2048
Entrenamiento	num_epochs	200
Entrenamiento	learning_rate	0.001
Entrenamiento	warmup_steps	1000
Entrenamiento	sparsity_warmup_steps	2000
Entrenamiento	sparsity_coef	0.3
P-Annealing	p_start	1.0
P-Annealing	p_end	0.2
P-Annealing	n_sparsity_updates	10
P-Annealing	anneal_start	5000
P-Annealing	estimated_epochs	200
Early Stopping	early_stopping	False
Early Stopping	patience	50

Archivo de activaciones: ..\..\..\activations\data\layer6_12500games.npy

Expansión: 16.0x

Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard_annealing_standard\12500games\ex3\plots

4. Dataset de Activaciones (Train/Validation Split 80-20%)

```
class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
```

```

        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})
print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')
print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [

```

```

        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False, # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)

```

```

=====
DATASET DE ACTIVACIONES
=====
      Propiedad      Valor
Total muestras      737,500
      Shape (737500, 512)
      Dtype      float32
Tamaño en memoria    1440.43 MB
=====

=====
SPLIT DE DATOS (80-20)
=====
      Conjunto Muestras Porcentaje
      Train    590,000      80.0%
Validation    147,500      20.0%
      Total    737,500     100.0%
=====

=====
DATALOADERS
=====
DataLoader  Batches  Batch Size Shuffle
      Train      289      2048      Sí
Validation      73      2048      No
=====

```

5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

Arquitectura:

- **Encoder:** Linear $\rightarrow$ ReLU (sparsity natural)- **Decoder:** Linear (reconstrucción)

```

# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)

```

```

# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim'] # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']} → {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']} → {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '-'
    ],
    'Parámetros': [
        '-',
        f"{encoder_params:,}",
        '-',
        f"{decoder_params:,}",
        '-',
        f"{num_params:,}"
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)

```

```

=====
ARQUITECTURA DEL MODELO

```

```

=====
Componente  Dimensión  Parámetros
Input      512         -
Encoder 512 → 8192  4,202,496
Hidden (Latent) 8192         -
Decoder 8192 → 512  4,194,304
Output     512         -
Total      -         8,397,312
=====

Factor de expansión: 16.0x
Tied weights: No
Device: cuda
=====

```

6. Función de Pérdida

Loss = MSE(reconstrucción) + λ × L1(activaciones)

- **MSE**: Mide qué tan bien reconstruimos las activaciones originales
- **L1**: Penaliza activaciones grandes → fuerza sparsity

```

class ConstrainedAdam(torch.optim.Adam):
    """
    Adam donde las columnas de decoder se proyectan a norma unitaria en cada step.
    Antes del paso: elimina la componente del gradiente paralela al decoder.
    Después del paso: renormaliza las columnas del decoder a norma 1.
    """
    def __init__(self, params, constrained_params, lr, betas=(0.9, 0.999)):
        super().__init__(params, lr=lr, betas=betas)
        self.constrained_params = list(constrained_params)

    def step(self, closure=None):
        with torch.no_grad():
            for p in self.constrained_params:
                normed_p = p / p.norm(dim=0, keepdim=True)
                p.grad -= (p.grad * normed_p).sum(dim=0, keepdim=True) * normed_p
        super().step(closure=closure)
        with torch.no_grad():
            for p in self.constrained_params:
                p /= p.norm(dim=0, keepdim=True)

```

```

def loss_function(x, reconstruction, hidden, sparsity_coeff, p, step=0, sparsity_warmup_steps=1000):
    """
    Pérdida compuesta: MSE (sum/mean) +  $L_p^p$  penalty con sparsity warmup

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse,  $\geq 0$  por ReLU)
        sparsity_coeff: Peso adaptativo de la penalización (alpha)
        p: Exponente actual del schedule de annealing
        step: Step global actual (para sparsity warmup)
        sparsity_warmup_steps: Steps de rampa de sparsity (0 = desactivado)
    """
    # BUG2 FIX: sum sobre features luego mean sobre batch (igual que referencia)
    mse_loss = (reconstruction - x).pow(2).sum(dim=-1).mean()
    # BUG4 FIX: sparsity warmup - rampa lineal de 0 a 1
    sparsity_scale = min(step / sparsity_warmup_steps, 1.0) if sparsity_warmup_steps > 0 else 1.0
    sparsity_loss = hidden.pow(p).sum(dim=1).mean() #  $L_p^p$ :  $\sum |f_i|^p$ 
    total_loss = mse_loss + sparsity_coeff * sparsity_scale * sparsity_loss
    return total_loss, mse_loss, sparsity_loss

# ConstrainedAdam: pasa model.decoder.parameters() como parámetros constrained
optimizer = ConstrainedAdam(
    model.parameters(),
    model.decoder.parameters(),
    lr=CONFIG['learning_rate']
)

# BUG3 FIX: LR warmup lineal - sube de 0 a lr en warmup_steps
def _lr_lambda(step):
    warmup = CONFIG.get('warmup_steps', 1000)
    return min(step / warmup, 1.0) if warmup > 0 else 1.0

scheduler = torch.optim.lr_scheduler.LambdaLR(optimizer, lr_lambda=_lr_lambda)

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'p_start', 'p_end'],
    'Valor': [
        'ConstrainedAdam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
        0,
        1
    ]
})

```

```

        f"{CONFIG['p_start']}",
        f"{CONFIG['p_end']}",
        f"{CONFIG['n_sparsity_updates']}",
        f"{CONFIG['anneal_start']}"
    ]
})

print('\n' + '='*55)
print(' OPTIMIZACIÓN')
print('='*55)
print(optim_info.to_string(index=False))
print('='*55)

```

```

=====
OPTIMIZACIÓN
=====

```

Parámetro	Valor
Optimizador	ConstrainedAdam
Learning Rate	1e-03
Sparsity Coef (inicial)	3e-01
p_start	1.0
p_end	0.2
n_sparsity_updates	10
anneal_start	5000

```

=====

```

7. Loop de Entrenamiento

```

# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],

```

```

    'val_fraction_active': [],
    'sparsity_coef_history': [],      # snapshot de sparsity_coef al final de cada época
    'p_history': [],                  # snapshot de p al final de cada época
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

# ===== P-ANNEALING STATE =====
p = CONFIG['p_start']
next_p = None
p_step_count = 0
sparsity_coef = CONFIG['sparsity_coef']
sparsity_queue = []
sparsity_queue_length = 10

total_steps = CONFIG['num_epochs'] * len(train_loader)
# BUG5 FIX: usar expected_steps (no total_steps) para que el schedule
# de annealing complete antes del early stopping
estimated_epochs = CONFIG.get('estimated_epochs', 50)
expected_steps = estimated_epochs * len(train_loader)
_update_steps_tensor = torch.linspace(
    CONFIG['anneal_start'], expected_steps, CONFIG['n_sparsity_updates'], dtype=torch.int
)
sparsity_update_steps_set = set(_update_steps_tensor.tolist())
sparsity_update_steps_list = sorted(sparsity_update_steps_set)
p_values = torch.linspace(CONFIG['p_start'], CONFIG['p_end'], CONFIG['n_sparsity_updates']).tolist()
global_step = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' P-annealing: p {CONFIG["p_start"]} → {CONFIG["p_end"]} en {CONFIG["n_sparsity_updates"]} épocas')
print(f' Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):

    # ===== ENTRENAMIENTO =====
    model.train()

    epoch_total_loss = 0

```

```

epoch_mse_loss = 0
epoch_sparsity_loss = 0
epoch_l0 = 0
epoch_fraction_active = 0

pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] p={p:.3f}')

for batch in pbar:
    x = batch.to(device)

    # Forward pass
    reconstruction, hidden = model(x)

    # ---- P-ANNEALING: lookahead para la cola ----
    with torch.no_grad():
        if next_p is not None:
            lp_curr = hidden.pow(p).sum(dim=1).mean().item()
            lp_next = hidden.pow(next_p).sum(dim=1).mean().item()
            sparsity_queue.append([lp_curr, lp_next])
            sparsity_queue = sparsity_queue[-sparsity_queue_length:]

    # Verificar si este step es un punto de actualización de p
    if (global_step in sparsity_update_steps_set
        and p_step_count < CONFIG['n_sparsity_updates']
        and global_step >= sparsity_update_steps_list[p_step_count]):
        # Re-escalar sparsity_coeff para compensar el cambio de magnitud de  $L_p^p$ 
        if next_p is not None and len(sparsity_queue) > 0:
            local_curr = sum(i[0] for i in sparsity_queue) / len(sparsity_queue)
            local_next = sum(i[1] for i in sparsity_queue) / len(sparsity_queue)
            if local_next > 0:
                sparsity_coeff = sparsity_coeff * (local_curr / local_next)
        # Actualizar p al siguiente valor del schedule
        p = p_values[p_step_count]
        if p_step_count < CONFIG['n_sparsity_updates'] - 1:
            next_p = p_values[p_step_count + 1]
        else:
            next_p = CONFIG['p_end']
        p_step_count += 1

    # Calcular pérdida con p actual
    total_loss, mse_loss, sparsity_loss = loss_function(
        x, reconstruction, hidden, sparsity_coeff, p,

```

```

        step=global_step,
        sparsity_warmup_steps=CONFIG.get('sparsity_warmup_steps', 0)
    )

    # Backward pass
    optimizer.zero_grad()
    total_loss.backward()
    optimizer.step()
    scheduler.step() # BUG3 FIX: LR warmup step

    global_step += 1

    # Métricas
    with torch.no_grad():
        l0 = (hidden > 0).float().sum(dim=1).mean().item()
        fraction_active = (hidden > 0).float().mean().item()

    epoch_total_loss += total_loss.item()
    epoch_mse_loss += mse_loss.item()
    epoch_sparsity_loss += sparsity_loss.item()
    epoch_l0 += l0
    epoch_fraction_active += fraction_active

    pbar.set_postfix({
        'loss': f'{total_loss.item():.4f}',
        'mse': f'{mse_loss.item():.4f}',
        'l0': f'{l0:.1f}',
        'p': f'{p:.3f}'
    })

    # Promedios de entrenamiento
    num_train_batches = len(train_loader)
    avg_train_total_loss = epoch_total_loss / num_train_batches
    avg_train_mse_loss = epoch_mse_loss / num_train_batches
    avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
    avg_train_l0 = epoch_l0 / num_train_batches
    avg_train_fraction_active = epoch_fraction_active / num_train_batches

    # ===== VALIDACIÓN =====
    model.eval()

    val_total_loss = 0

```

```

val_mse_loss = 0
val_sparsity_loss = 0
val_l0 = 0
val_fraction_active = 0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        reconstruction, hidden = model(x)
        total_loss, mse_loss, sparsity_loss = loss_function(
            x, reconstruction, hidden, sparsity_coeff, p,
            step=global_step,
            sparsity_warmup_steps=CONFIG.get('sparsity_warmup_steps', 0)
        )
        l0 = (hidden > 0).float().sum(dim=1).mean().item()
        fraction_active = (hidden > 0).float().mean().item()

        val_total_loss += total_loss.item()
        val_mse_loss += mse_loss.item()
        val_sparsity_loss += sparsity_loss.item()
        val_l0 += l0
        val_fraction_active += fraction_active

num_val_batches = len(val_loader)
avg_val_total_loss = val_total_loss / num_val_batches
avg_val_mse_loss = val_mse_loss / num_val_batches
avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
avg_val_l0 = val_l0 / num_val_batches
avg_val_fraction_active = val_fraction_active / num_val_batches

# Guardar en historial
history['train_total_loss'].append(avg_train_total_loss)
history['train_mse_loss'].append(avg_train_mse_loss)
history['train_sparsity_loss'].append(avg_train_sparsity_loss)
history['train_l0_sparsity'].append(avg_train_l0)
history['train_fraction_active'].append(avg_train_fraction_active)

history['val_total_loss'].append(avg_val_total_loss)
history['val_mse_loss'].append(avg_val_mse_loss)
history['val_sparsity_loss'].append(avg_val_sparsity_loss)
history['val_l0_sparsity'].append(avg_val_l0)
history['val_fraction_active'].append(avg_val_fraction_active)

```

```

history['sparsity_coef_history'].append(sparsity_coef)
history['p_history'].append(p)

# ===== EARLY STOPPING =====
if CONFIG['early_stopping']:
    if avg_val_mse_loss < best_val_mse:
        best_val_mse = avg_val_mse_loss
        best_epoch = epoch + 1
        epochs_no_improve = 0

    best_checkpoint = {
        'model_state_dict': model.state_dict(),
        'optimizer_state_dict': optimizer.state_dict(),
        'config': CONFIG,
        'history': history,
        'epoch': best_epoch,
        'val_mse': best_val_mse
    }

    best_model_name = get_model_name(
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best'
    )
    torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='best',
    extras=NAMING_META['extras'],
    config=CONFIG,
    history=history,
    optimizer=optimizer
)

```

```

        print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f}')
    else:
        epochs_no_improve += 1
        if epochs_no_improve >= CONFIG['patience']:
            print(f'\n Early stopping activado en época {epoch+1}')
            print(f' Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}')
            break

# Log cada 5 épocas
if (epoch + 1) % 5 == 0 or epoch == 0:
    print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | p: {p:.3f} | : {sparsity_coeff:.3f}')
    print(f' Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}')
    print(f' Val - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0: {l0_sparsity:.3f}')
    if CONFIG['early_stopping']:
        print(f' Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {epochs_no_improve}')

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})

```

```
print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)
```

Iniciando entrenamiento: 200 épocas máximo
P-annealing: p 1.0 → 0.2 en 10 saltos
Early stopping: patience=50 épocas

```
=====
```

Época 1/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:11<00:00, 24.98it/s, loss=19.73]

Época 1/200 | p: 1.000 | : 3.00e-01
Train - Total: 18.5541 | MSE: 10.1224 | L0: 4846.5
Val - Total: 19.7317 | MSE: 4.4647 | L0: 4627.0

Época 2/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:11<00:00, 25.06it/s, loss=28.55]
Época 3/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:11<00:00, 26.17it/s, loss=37.60]
Época 4/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:12<00:00, 23.40it/s, loss=43.61]
Época 5/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:10<00:00, 27.78it/s, loss=59.79]

Época 5/200 | p: 1.000 | : 3.00e-01
Train - Total: 46.6758 | MSE: 12.5411 | L0: 1727.7
Val - Total: 50.2713 | MSE: 13.7496 | L0: 1589.5

Época 6/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:10<00:00, 27.86it/s, loss=55.79]
Época 7/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:10<00:00, 27.91it/s, loss=60.60]
Época 8/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:10<00:00, 27.30it/s, loss=61.61]
Época 9/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:10<00:00, 27.94it/s, loss=59.79]
Época 10/200 [Train] p=1.000 =3.00e-01: 100%| | 289/289 [00:10<00:00, 27.91it/s, loss=59.79]

Época 10/200 | p: 1.000 | : 3.00e-01
Train - Total: 59.7139 | MSE: 18.0242 | L0: 1038.9
Val - Total: 60.6744 | MSE: 18.7035 | L0: 1048.1

Época 11/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:12<00:00,	23.81it/s,	loss=58
Época 12/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:11<00:00,	25.18it/s,	loss=60
Época 13/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:13<00:00,	22.14it/s,	loss=58
Época 14/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:12<00:00,	22.83it/s,	loss=60
Época 15/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:12<00:00,	22.52it/s,	loss=58

Época 15/200 | p: 1.000 | : 3.00e-01
 Train - Total: 58.7912 | MSE: 18.1480 | L0: 958.6
 Val - Total: 60.0092 | MSE: 19.1311 | L0: 973.2

Época 16/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:12<00:00,	22.75it/s,	loss=58
Época 17/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:12<00:00,	22.46it/s,	loss=58
Época 18/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:12<00:00,	22.24it/s,	loss=57
Época 19/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:13<00:00,	21.24it/s,	loss=57
Época 20/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:12<00:00,	23.22it/s,	loss=58

Época 20/200 | p: 1.000 | : 3.00e-01
 Train - Total: 58.3151 | MSE: 18.1755 | L0: 920.0
 Val - Total: 59.6817 | MSE: 19.6347 | L0: 928.4

Época 21/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:12<00:00,	22.40it/s,	loss=58
Época 22/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	26.47it/s,	loss=58
Época 23/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	27.05it/s,	loss=60
Época 24/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	27.21it/s,	loss=58
Época 25/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	27.10it/s,	loss=58

Época 25/200 | p: 1.000 | : 3.00e-01
 Train - Total: 58.0709 | MSE: 18.3235 | L0: 888.4
 Val - Total: 59.5276 | MSE: 19.6939 | L0: 899.8

Época 26/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	27.34it/s,	loss=58
Época 27/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	27.28it/s,	loss=58
Época 28/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	27.23it/s,	loss=57
Época 29/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	27.15it/s,	loss=57
Época 30/200 [Train]	p=1.000	=3.00e-01:	100%		289/289	[00:10<00:00,	27.20it/s,	loss=58

Época 30/200 | p: 1.000 | : 3.00e-01
Train - Total: 57.9223 | MSE: 18.4317 | L0: 867.0
Val - Total: 59.4521 | MSE: 19.9086 | L0: 877.8

Época 31/200 [Train] p=1.000 =3.00e-01: 100%	289/289 [00:10<00:00, 27.28it/s, loss=58
Época 32/200 [Train] p=1.000 =3.00e-01: 100%	289/289 [00:10<00:00, 27.08it/s, loss=58
Época 33/200 [Train] p=1.000 =3.00e-01: 100%	289/289 [00:10<00:00, 27.41it/s, loss=58
Época 34/200 [Train] p=1.000 =3.00e-01: 100%	289/289 [00:10<00:00, 27.15it/s, loss=58
Época 35/200 [Train] p=1.000 =3.00e-01: 100%	289/289 [00:10<00:00, 27.31it/s, loss=57

Época 35/200 | p: 1.000 | : 3.00e-01
Train - Total: 57.8161 | MSE: 18.4918 | L0: 854.2
Val - Total: 59.3948 | MSE: 20.0229 | L0: 865.7

Época 36/200 [Train] p=1.000 =3.00e-01: 100%	289/289 [00:10<00:00, 27.14it/s, loss=58
Época 37/200 [Train] p=1.000 =3.00e-01: 100%	289/289 [00:10<00:00, 27.37it/s, loss=57
Época 38/200 [Train] p=1.000 =3.00e-01: 100%	289/289 [00:10<00:00, 27.04it/s, loss=57
Época 39/200 [Train] p=0.911 =2.65e-01: 100%	289/289 [00:10<00:00, 27.23it/s, loss=58
Época 40/200 [Train] p=0.911 =2.65e-01: 100%	289/289 [00:10<00:00, 26.93it/s, loss=57

Época 40/200 | p: 0.911 | : 2.65e-01
Train - Total: 57.7747 | MSE: 19.3738 | L0: 759.7
Val - Total: 59.5332 | MSE: 21.2140 | L0: 761.3

Época 41/200 [Train] p=0.911 =2.65e-01: 100%	289/289 [00:10<00:00, 27.30it/s, loss=57
Época 42/200 [Train] p=0.911 =2.65e-01: 100%	289/289 [00:10<00:00, 26.85it/s, loss=57
Época 43/200 [Train] p=0.911 =2.65e-01: 100%	289/289 [00:10<00:00, 27.26it/s, loss=59
Época 44/200 [Train] p=0.911 =2.65e-01: 100%	289/289 [00:10<00:00, 26.99it/s, loss=57
Época 45/200 [Train] p=0.911 =2.65e-01: 100%	289/289 [00:10<00:00, 27.08it/s, loss=58

Época 45/200 | p: 0.911 | : 2.65e-01
Train - Total: 57.7107 | MSE: 19.5935 | L0: 723.6
Val - Total: 59.5412 | MSE: 21.5473 | L0: 729.1

Época 46/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.23it/s,	loss=57
Época 47/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.10it/s,	loss=58
Época 48/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.19it/s,	loss=58
Época 49/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.06it/s,	loss=56
Época 50/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	26.96it/s,	loss=57

Época 50/200 | p: 0.911 | : 2.65e-01
 Train - Total: 57.6221 | MSE: 19.5789 | L0: 709.3
 Val - Total: 59.5025 | MSE: 21.4729 | L0: 716.9

Época 51/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.03it/s,	loss=57
Época 52/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.20it/s,	loss=56
Época 53/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.02it/s,	loss=57
Época 54/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.06it/s,	loss=60
Época 55/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.11it/s,	loss=59

Época 55/200 | p: 0.911 | : 2.65e-01
 Train - Total: 57.5807 | MSE: 19.5500 | L0: 700.0
 Val - Total: 59.4415 | MSE: 21.4500 | L0: 707.6

Época 56/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	26.80it/s,	loss=58
Época 57/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.03it/s,	loss=57
Época 58/200 [Train]	p=0.911	=2.65e-01:	100%		289/289	[00:10<00:00,	27.13it/s,	loss=58
Época 59/200 [Train]	p=0.822	=2.37e-01:	100%		289/289	[00:10<00:00,	27.07it/s,	loss=57
Época 60/200 [Train]	p=0.822	=2.37e-01:	100%		289/289	[00:10<00:00,	27.27it/s,	loss=57

Época 60/200 | p: 0.822 | : 2.37e-01
 Train - Total: 57.4778 | MSE: 20.0474 | L0: 635.5
 Val - Total: 59.4659 | MSE: 22.0853 | L0: 635.4

Época 61/200 [Train]	p=0.822	=2.37e-01:	100%		289/289	[00:10<00:00,	27.05it/s,	loss=57
Época 62/200 [Train]	p=0.822	=2.37e-01:	100%		289/289	[00:10<00:00,	27.25it/s,	loss=58
Época 63/200 [Train]	p=0.822	=2.37e-01:	100%		289/289	[00:10<00:00,	26.99it/s,	loss=58
Época 64/200 [Train]	p=0.822	=2.37e-01:	100%		289/289	[00:10<00:00,	27.38it/s,	loss=58
Época 65/200 [Train]	p=0.822	=2.37e-01:	100%		289/289	[00:10<00:00,	27.06it/s,	loss=57

Época 65/200 | p: 0.822 | : 2.37e-01
Train - Total: 57.3224 | MSE: 19.9492 | L0: 599.9
Val - Total: 59.3241 | MSE: 21.8768 | L0: 605.5

Época 66/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.28it/s, loss=57
Época 67/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.21it/s, loss=57
Época 68/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.19it/s, loss=56
Época 69/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.36it/s, loss=56
Época 70/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.00it/s, loss=57

Época 70/200 | p: 0.822 | : 2.37e-01
Train - Total: 57.1445 | MSE: 19.6005 | L0: 583.9
Val - Total: 59.1393 | MSE: 21.5504 | L0: 589.4

Época 71/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.23it/s, loss=56
Época 72/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.02it/s, loss=57
Época 73/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.24it/s, loss=57
Época 74/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.09it/s, loss=59
Época 75/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.18it/s, loss=56

Época 75/200 | p: 0.822 | : 2.37e-01
Train - Total: 56.9963 | MSE: 19.2447 | L0: 572.1
Val - Total: 58.9877 | MSE: 21.2189 | L0: 577.4

Época 76/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.20it/s, loss=56
Época 77/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.25it/s, loss=58
Época 78/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.13it/s, loss=56
Época 79/200 [Train] p=0.822 =2.37e-01: 100%| | 289/289 [00:10<00:00, 27.02it/s, loss=57
Época 80/200 [Train] p=0.733 =2.16e-01: 100%| | 289/289 [00:10<00:00, 26.89it/s, loss=57

Época 80/200 | p: 0.733 | : 2.16e-01
Train - Total: 56.7941 | MSE: 19.2825 | L0: 525.2
Val - Total: 58.7638 | MSE: 21.2794 | L0: 523.6

Época 81/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.04it/s, loss=57.00]
Época 82/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.08it/s, loss=57.00]
Época 83/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.00it/s, loss=57.00]
Época 84/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.10it/s, loss=57.00]
Época 85/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.12it/s, loss=57.00]

Época 85/200 | p: 0.733 | : 2.16e-01
 Train - Total: 56.3754 | MSE: 18.7942 | L0: 491.2
 Val - Total: 58.3639 | MSE: 20.9206 | L0: 491.5

Época 86/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.11it/s, loss=57.00]
Época 87/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.00it/s, loss=57.00]
Época 88/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.09it/s, loss=57.00]
Época 89/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.11it/s, loss=57.00]
Época 90/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.07it/s, loss=57.00]

Época 90/200 | p: 0.733 | : 2.16e-01
 Train - Total: 56.0058 | MSE: 18.1621 | L0: 475.9
 Val - Total: 57.9639 | MSE: 20.0902 | L0: 479.0

Época 91/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 26.89it/s, loss=57.00]
Época 92/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 26.96it/s, loss=57.00]
Época 93/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.01it/s, loss=57.00]
Época 94/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.06it/s, loss=57.00]
Época 95/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 26.92it/s, loss=57.00]

Época 95/200 | p: 0.733 | : 2.16e-01
 Train - Total: 55.7368 | MSE: 17.7466 | L0: 465.6
 Val - Total: 57.7417 | MSE: 19.8420 | L0: 467.4

Época 96/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 26.93it/s, loss=57.00]
Época 97/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.08it/s, loss=57.00]
Época 98/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 27.04it/s, loss=57.00]
Época 99/200 [Train] p=0.733 =2.16e-01: 100%	289/289 [00:10<00:00, 26.86it/s, loss=57.00]
Época 100/200 [Train] p=0.644 =2.01e-01: 100%	289/289 [00:10<00:00, 26.98it/s, loss=57.00]

Época 100/200 | p: 0.644 | : 2.01e-01
Train - Total: 55.7803 | MSE: 18.1694 | L0: 436.1
Val - Total: 57.8026 | MSE: 20.3923 | L0: 432.5

Época 101/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.13it/s, loss=5
Época 102/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.08it/s, loss=5
Época 103/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.07it/s, loss=5
Época 104/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.12it/s, loss=5
Época 105/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 26.97it/s, loss=5

Época 105/200 | p: 0.644 | : 2.01e-01
Train - Total: 55.4063 | MSE: 18.1274 | L0: 410.9
Val - Total: 57.5868 | MSE: 20.5237 | L0: 410.4

Época 106/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.10it/s, loss=5
Época 107/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.04it/s, loss=5
Época 108/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.14it/s, loss=5
Época 109/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.07it/s, loss=5
Época 110/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.06it/s, loss=5

Época 110/200 | p: 0.644 | : 2.01e-01
Train - Total: 55.1858 | MSE: 17.9501 | L0: 402.6
Val - Total: 57.4406 | MSE: 20.3021 | L0: 404.2

Época 111/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.10it/s, loss=5
Época 112/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.03it/s, loss=5
Época 113/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.07it/s, loss=5
Época 114/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.07it/s, loss=5
Época 115/200 [Train] p=0.644 =2.01e-01: 100%| | 289/289 [00:10<00:00, 27.10it/s, loss=5

Época 115/200 | p: 0.644 | : 2.01e-01
Train - Total: 55.0676 | MSE: 17.8640 | L0: 397.7
Val - Total: 57.3961 | MSE: 20.3555 | L0: 398.8

Época 116/200 [Train]	p=0.644	=2.01e-01: 100%	289/289 [00:10<00:00, 27.04it/s, loss=5
Época 117/200 [Train]	p=0.644	=2.01e-01: 100%	289/289 [00:10<00:00, 26.99it/s, loss=5
Época 118/200 [Train]	p=0.644	=2.01e-01: 100%	289/289 [00:10<00:00, 27.04it/s, loss=5
Época 119/200 [Train]	p=0.644	=2.01e-01: 100%	289/289 [00:10<00:00, 26.87it/s, loss=5
Época 120/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.07it/s, loss=5

Época 120/200 | p: 0.556 | : 1.87e-01
 Train - Total: 55.5981 | MSE: 18.8292 | L0: 380.8
 Val - Total: 57.9255 | MSE: 21.4823 | L0: 376.3

Época 121/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 26.95it/s, loss=5
Época 122/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.17it/s, loss=5
Época 123/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.20it/s, loss=5
Época 124/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.06it/s, loss=5
Época 125/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.03it/s, loss=5

Época 125/200 | p: 0.556 | : 1.87e-01
 Train - Total: 55.2349 | MSE: 19.0270 | L0: 361.2
 Val - Total: 57.7764 | MSE: 21.7051 | L0: 361.9

Época 126/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.03it/s, loss=5
Época 127/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.14it/s, loss=5
Época 128/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.23it/s, loss=5
Época 129/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.02it/s, loss=5
Época 130/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.07it/s, loss=5

Época 130/200 | p: 0.556 | : 1.87e-01
 Train - Total: 55.1034 | MSE: 18.9796 | L0: 356.4
 Val - Total: 57.7346 | MSE: 21.7516 | L0: 357.2

Época 131/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.06it/s, loss=5
Época 132/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.14it/s, loss=5
Época 133/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.16it/s, loss=5
Época 134/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.04it/s, loss=5
Época 135/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.10it/s, loss=5

Época 135/200 | p: 0.556 | : 1.87e-01
Train - Total: 54.9777 | MSE: 18.8627 | L0: 353.8
Val - Total: 57.6207 | MSE: 21.7125 | L0: 353.8

Época 136/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 26.97it/s, loss=5
Época 137/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.19it/s, loss=5
Época 138/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.20it/s, loss=5
Época 139/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.04it/s, loss=5
Época 140/200 [Train]	p=0.556	=1.87e-01: 100%	289/289 [00:10<00:00, 27.02it/s, loss=5

Época 140/200 | p: 0.467 | : 1.75e-01
Train - Total: 55.8668 | MSE: 20.0741 | L0: 344.4
Val - Total: 58.7237 | MSE: 23.3025 | L0: 339.3

Época 141/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 27.08it/s, loss=5
Época 142/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 27.27it/s, loss=5
Época 143/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 27.02it/s, loss=5
Época 144/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 27.03it/s, loss=5
Época 145/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 26.83it/s, loss=5

Época 145/200 | p: 0.467 | : 1.75e-01
Train - Total: 55.8434 | MSE: 20.7401 | L0: 326.7
Val - Total: 58.5037 | MSE: 23.5358 | L0: 326.8

Época 146/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 27.04it/s, loss=5
Época 147/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 27.03it/s, loss=5
Época 148/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 26.92it/s, loss=5
Época 149/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 27.06it/s, loss=5
Época 150/200 [Train]	p=0.467	=1.75e-01: 100%	289/289 [00:10<00:00, 27.00it/s, loss=5

Época 150/200 | p: 0.467 | : 1.75e-01
Train - Total: 55.5848 | MSE: 20.5225 | L0: 323.8
Val - Total: 58.2800 | MSE: 23.3365 | L0: 324.0

Época 151/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 27.12it/s, loss=5
Época 152/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 27.10it/s, loss=5
Época 153/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 27.09it/s, loss=5
Época 154/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 27.05it/s, loss=5
Época 155/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 27.11it/s, loss=5

Época 155/200 | p: 0.467 | : 1.75e-01
 Train - Total: 55.4312 | MSE: 20.3138 | L0: 322.4
 Val - Total: 58.1858 | MSE: 23.2707 | L0: 321.8

Época 156/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 27.04it/s, loss=5
Época 157/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 26.98it/s, loss=5
Época 158/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 26.93it/s, loss=5
Época 159/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 27.18it/s, loss=5
Época 160/200 [Train]	p=0.467	=1.75e-01: 100%		289/289	[00:10<00:00, 26.97it/s, loss=5

Época 160/200 | p: 0.378 | : 1.64e-01
 Train - Total: 56.0498 | MSE: 21.0073 | L0: 318.7
 Val - Total: 59.6479 | MSE: 24.9974 | L0: 313.1

Época 161/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 27.16it/s, loss=5
Época 162/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 27.03it/s, loss=5
Época 163/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 27.00it/s, loss=5
Época 164/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 27.09it/s, loss=5
Época 165/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 27.04it/s, loss=5

Época 165/200 | p: 0.378 | : 1.64e-01
 Train - Total: 57.0516 | MSE: 22.6889 | L0: 303.8
 Val - Total: 59.5237 | MSE: 25.2028 | L0: 304.0

Época 166/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 27.13it/s, loss=5
Época 167/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 27.07it/s, loss=5
Época 168/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 26.93it/s, loss=5
Época 169/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 27.02it/s, loss=5
Época 170/200 [Train]	p=0.378	=1.64e-01: 100%		289/289	[00:10<00:00, 26.99it/s, loss=5

Época 170/200 | p: 0.378 | : 1.64e-01
Train - Total: 56.7383 | MSE: 22.2689 | L0: 302.5
Val - Total: 59.2372 | MSE: 24.8243 | L0: 302.8

Época 171/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 26.99it/s, loss=5
Época 172/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.10it/s, loss=5
Época 173/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.04it/s, loss=5
Época 174/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.10it/s, loss=5
Época 175/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.03it/s, loss=5

Época 175/200 | p: 0.378 | : 1.64e-01
Train - Total: 56.4460 | MSE: 21.8297 | L0: 302.6
Val - Total: 58.8639 | MSE: 24.3099 | L0: 302.8

Época 176/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.10it/s, loss=5
Época 177/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.19it/s, loss=5
Época 178/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.09it/s, loss=5
Época 179/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.17it/s, loss=5
Época 180/200 [Train]	p=0.378	=1.64e-01: 100%	289/289	[00:10<00:00, 27.08it/s, loss=5

Época 180/200 | p: 0.289 | : 1.53e-01
Train - Total: 56.5426 | MSE: 21.8075 | L0: 302.2
Val - Total: 60.2848 | MSE: 25.7335 | L0: 299.3

Época 181/200 [Train]	p=0.289	=1.53e-01: 100%	289/289	[00:10<00:00, 27.07it/s, loss=5
Época 182/200 [Train]	p=0.289	=1.53e-01: 100%	289/289	[00:10<00:00, 27.13it/s, loss=5
Época 183/200 [Train]	p=0.289	=1.53e-01: 100%	289/289	[00:10<00:00, 26.95it/s, loss=5
Época 184/200 [Train]	p=0.289	=1.53e-01: 100%	289/289	[00:10<00:00, 27.03it/s, loss=6
Época 185/200 [Train]	p=0.289	=1.53e-01: 100%	289/289	[00:10<00:00, 26.99it/s, loss=6

Época 185/200 | p: 0.289 | : 1.53e-01
Train - Total: 58.8618 | MSE: 24.6600 | L0: 290.8
Val - Total: 60.9761 | MSE: 26.8062 | L0: 290.6

```

Época 186/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.09it/s, loss=5
Época 187/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.10it/s, loss=5
Época 188/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.15it/s, loss=5
Época 189/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.18it/s, loss=5
Época 190/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.01it/s, loss=5

```

```

Época 190/200 | p: 0.289 | : 1.53e-01
Train - Total: 58.6603 | MSE: 24.5059 | L0: 289.1
Val   - Total: 60.9793 | MSE: 26.8474 | L0: 289.2

```

```

Época 191/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 26.99it/s, loss=5
Época 192/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 26.95it/s, loss=5
Época 193/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.20it/s, loss=5
Época 194/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 26.92it/s, loss=5
Época 195/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 26.99it/s, loss=5

```

```

Época 195/200 | p: 0.289 | : 1.53e-01
Train - Total: 58.4343 | MSE: 24.2600 | L0: 288.6
Val   - Total: 60.6666 | MSE: 26.6289 | L0: 287.9

```

```

Época 196/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 26.97it/s, loss=5
Época 197/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.01it/s, loss=5
Época 198/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.00it/s, loss=5
Época 199/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.00it/s, loss=5
Época 200/200 [Train] p=0.289 =1.53e-01: 100%| | 289/289 [00:10<00:00, 27.02it/s, loss=5

```

```

Época 200/200 | p: 0.289 | : 1.53e-01
Train - Total: 58.1165 | MSE: 23.8850 | L0: 288.9
Val   - Total: 60.3601 | MSE: 26.2374 | L0: 288.4

```

===== RESUMEN DEL ENTRENAMIENTO =====

Entrenamiento completo: 200 épocas

Métricas Finales (Última Época):

Métrica	Train	Validation
Total Loss	58.1165	60.3601

=====

```

    estado='final',
    extras=NAMING_META['extras'],
    config=CONFIG,
    history=history,
    optimizer=optimizer
)

# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)

```

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard_annealing_standard\12500games\sae_standard_p-annealing_l6_12500g_ex3_final.pt

```

=====
CHECKPOINTS GUARDADOS
=====

```

Tipo	Ubicación
Local	saved_model\sae_standard_p-annealing_l6_12500g_final.pt
Organizado	layer_06/models/...

```

=====

```

9. Métricas básicas del modelo

```

# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],

```

```

NAMING_META["capa"],
NAMING_META["juegos"],
extras_id=extras_id
)

training_metrics = {
    # Metricas de calidad
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "p_start": CONFIG["p_start"],
    "p_end": CONFIG["p_end"],
    "n_sparsity_updates": CONFIG["n_sparsity_updates"],
    "anneal_start": CONFIG["anneal_start"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}

metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",
        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})
print("" + "="*50)

```

```

print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)

```

```

=====
METRICAS DE ENTRENAMIENTO GUARDADAS
=====

```

	Metrica	Valor
Val MSE (final)		26.237351
Val MSE (mejor)		inf
Val LO (final)		288.4
Fraccion Activa		3.52%
Archivo:	c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\annealing_standard\12500games\ex3\metrics\training_metrics.json	

```

=====

```

10. Visualización de Resultados

```

# Gráficas de pérdidas (Train vs Validation) + P-annealing schedule
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')
axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best')
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')

```

```

axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8)
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0.8)
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# P schedule (annealing)
axes[1, 1].plot(history['p_history'], color='brown', linewidth=2, label='p')
axes[1, 1].set_title('P Schedule (Annealing)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('p')
axes[1, 1].set_ylim(0, 1.1)
axes[1, 1].grid(True, alpha=0.3)

# Train vs Val MSE Gap
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best')
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

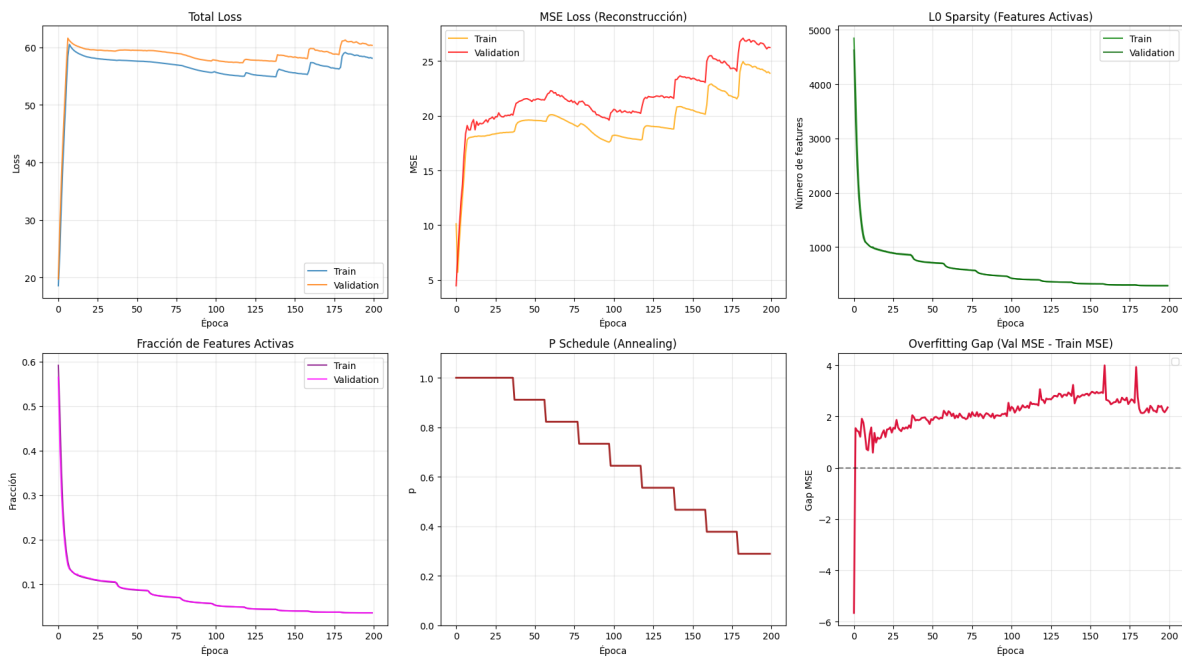
plt.tight_layout()

```

```
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('  Gráficas guardadas exitosamente')
print('='*50)
```

C:\Users\Esposa\AppData\Local\Temp\ipykernel_28044\2801488273.py:59: UserWarning: No artists axes[1, 2].legend()



```
=====
Gráficas guardadas exitosamente
=====
```

11. Resumen Final

```
# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
```

```

print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:}"
    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):}",
        f"{len(val_dataset):}",
        f"{len(full_dataset):}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')
print(final_summary.to_string(index=False))

# Archivos

```

```

files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))

print('\n' + '='*70)
print('  ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)

```

```

=====
RESUMEN FINAL DEL ENTRENAMIENTO
=====

```

Arquitectura:

Componente	Valor
Input	512
Hidden	8192 (16.0x)
Parámetros	8,397,312

Datos:

Conjunto Muestras	
Train	590,000
Validation	147,500
Total	737,500

Métricas Finales:

Métrica	Valor
Train Loss	58.1165
Val Loss	60.3601
Train MSE	23.8850
Val MSE	26.2374
Train L0	288.9
Val L0	288.4

Archivos Guardados:

Tipo

```
Modelo
annealing_16_12500g_final.pt
Gráficas c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard
annealing_standard\12500games\ex3\plots\training_curves.png
```

```
=====
ENTRENAMIENTO COMPLETADO EXITOSAMENTE
=====
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_standard_pannealing.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
```

```

# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard_p-annealing_l6_12500g_ex3_training.pdf

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard_p-annealing_l6_12500g_ex3_training.pdf